

# Machine Learning Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence

## Executive Summary

The supplied data contains 2,000 client records and 10,000 property records. Client demographics were joined to sold-property behavior and clustered with K-Means. Four clusters are used as the final business segmentation. The K=4 silhouette score is 0.1510, while hierarchical clustering at K=4 scores 0.1351; therefore the groups should be treated as descriptive analytical segments rather than perfectly separated natural classes.

## Dataset

clients.csv has 2,000 rows and 14 columns. properties.csv has 10,000 rows and 11 columns. The property date range is 2024-01-01 to 2025-01-12. There are 7,305 sold and 2,695 available records.

## Cleaning and Feature Engineering

Currency values were converted to numeric; dates were parsed; mixed DOB formats were normalized; age was calculated; sold properties were aggregated by client; apartment/office mix and price per square foot were derived; categorical features were one-hot encoded and numeric features standardized.

## Modeling

K-Means was evaluated for K=2 through K=8 using inertia and silhouette score. Hierarchical clustering with Ward linkage was used as validation. K=4 was selected for the final business segmentation to align with the PRD's four-segment structure while transparently reporting the moderate clustering separation.

## Business Interpretation

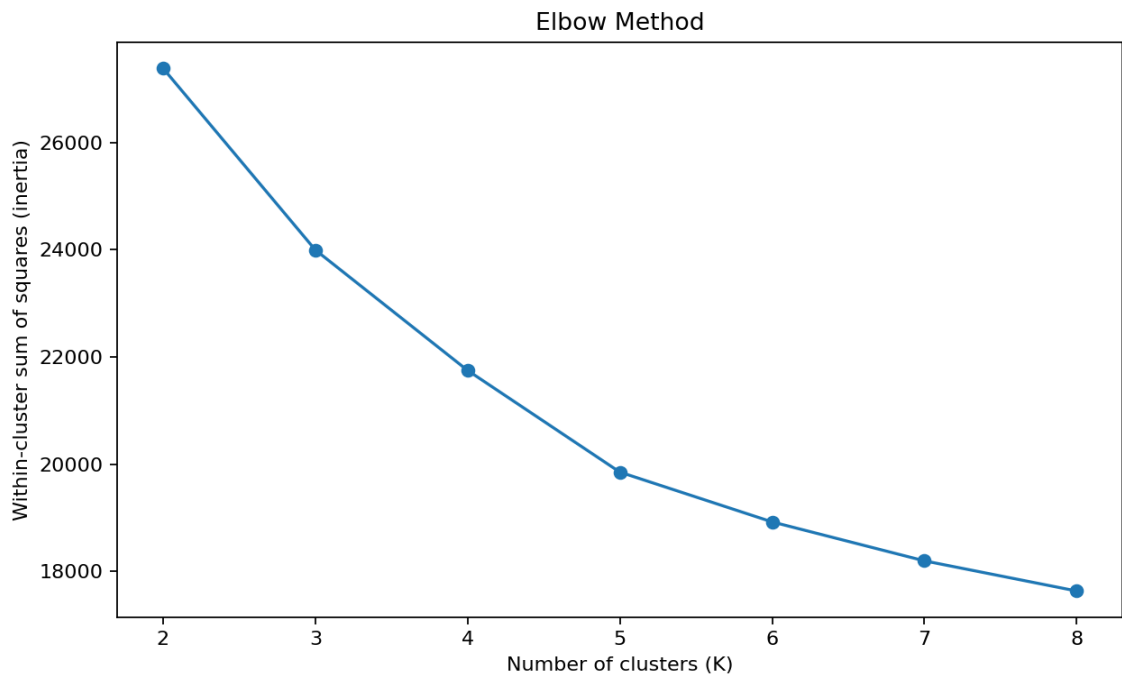
The final segments are descriptive labels based on observed purchase count, purchase value, property mix, demographics, financing, geography and satisfaction. No unsupported income or wealth claims are made because those fields are not present in the supplied dataset.

## Dashboard

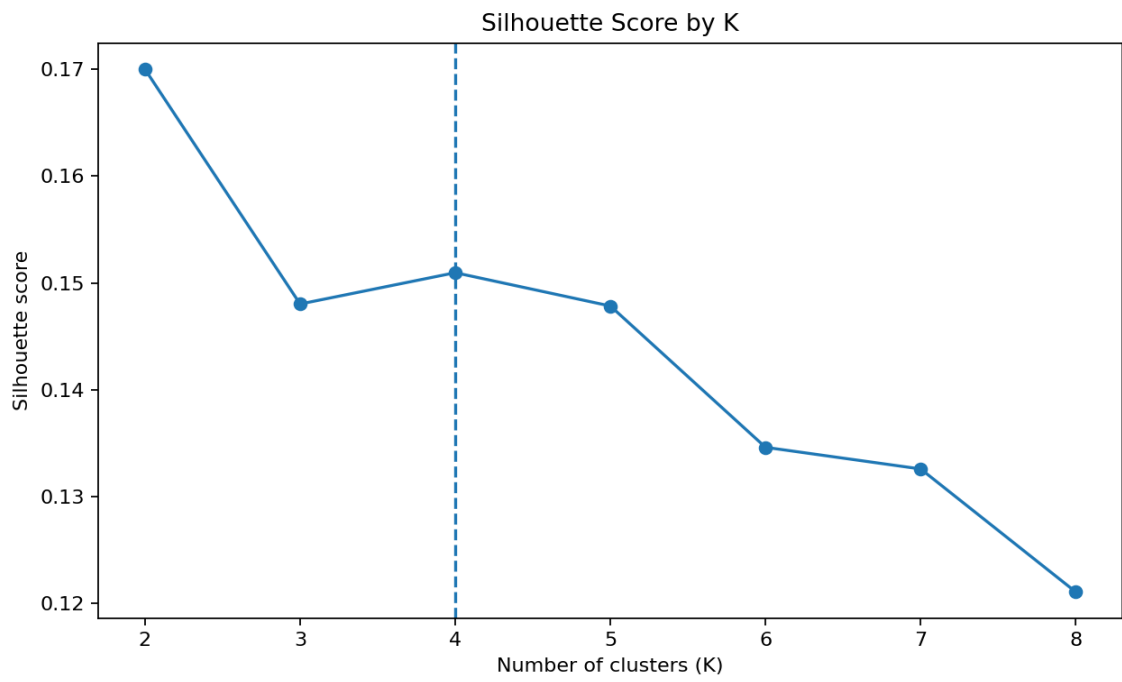
The Streamlit application provides country, region, acquisition-purpose and client-type filters, plus cluster distribution, investor behavior, geographic analysis, segment profiles and clustering validation.

## Cluster Summary

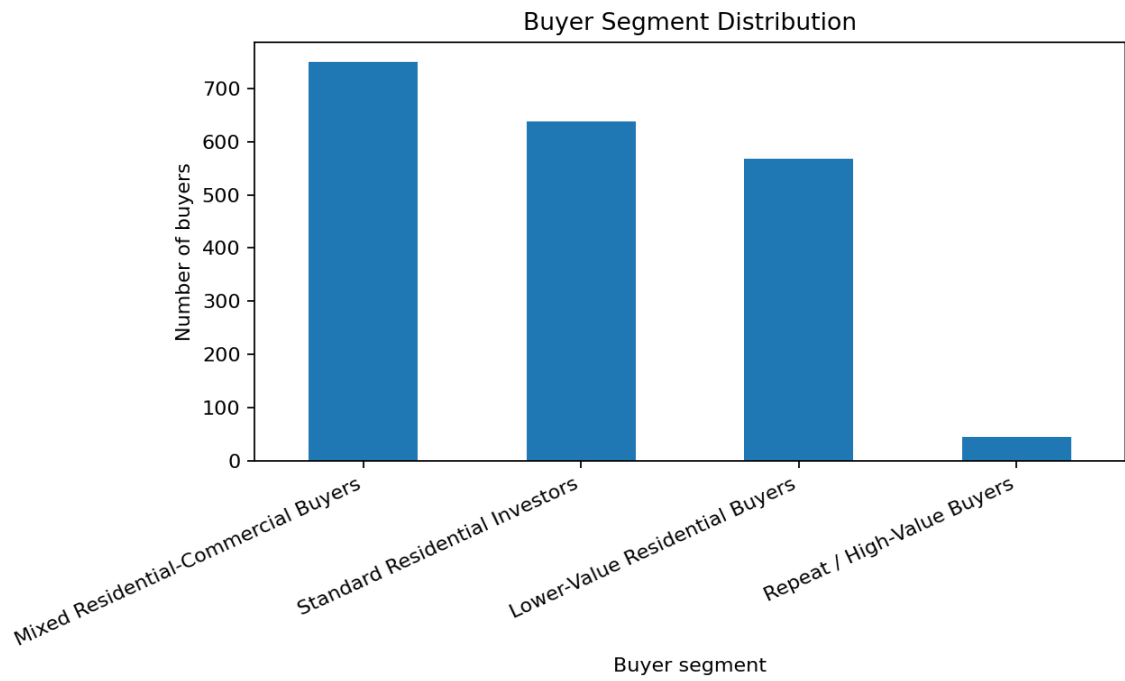
| Cluster | Segment                             | Buyers | Avg Age | Avg Properties | Avg Total Value | Investment % |
|---------|-------------------------------------|--------|---------|----------------|-----------------|--------------|
| 0       | Repeat / High-Value Buyers          | 46     | 62.5    | 7.43           | \$2,476,739     | 34.8%        |
| 1       | Standard Residential Investors      | 638    | 54.7    | 3.64           | \$1,395,722     | 28.4%        |
| 2       | Lower-Value Residential Buyers      | 567    | 54.5    | 3.40           | \$941,317       | 33.2%        |
| 3       | Mixed Residential-Commercial Buyers | 349    | 54.2    | 3.62           | \$1,311,914     | 30.7%        |



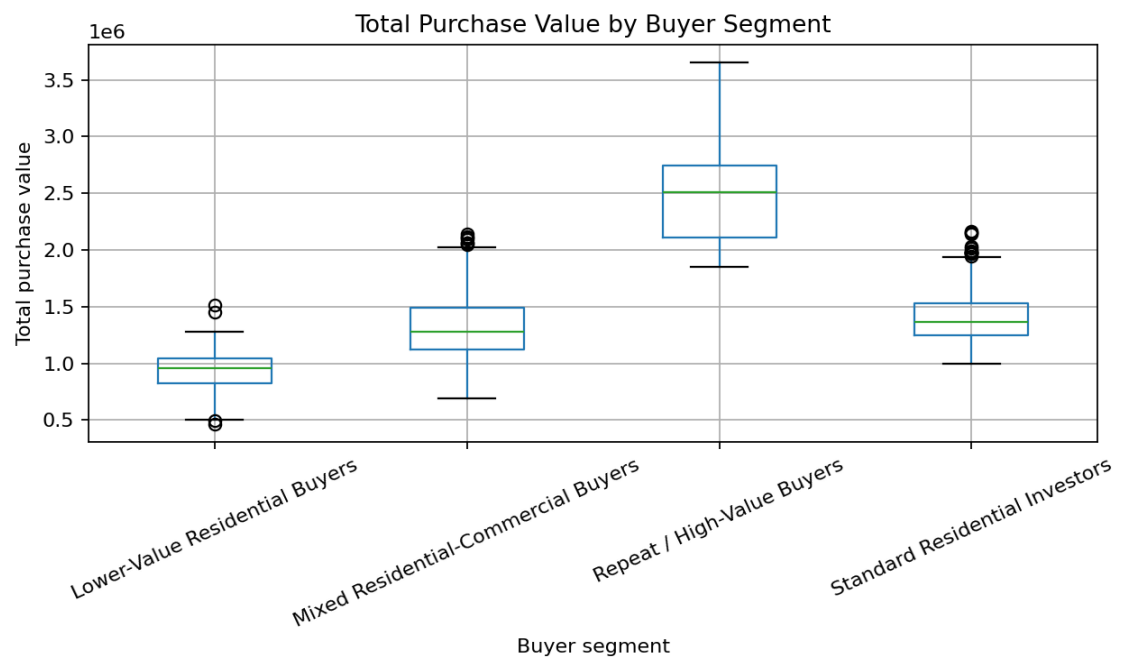
Elbow Method



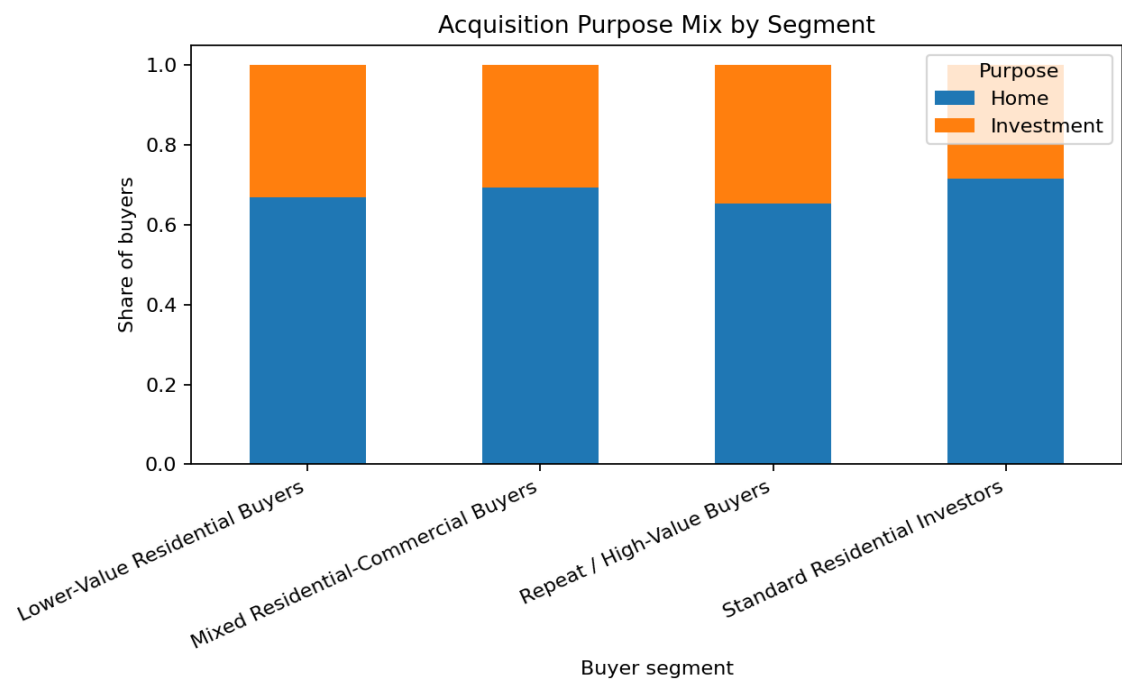
Silhouette Scores



Buyer Segment Distribution



Purchase Value by Segment



Acquisition Purpose Mix